Individual Assignment

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Overview

Basic Information

- Total Number of records/rows/lines: 260,503 records
- Total Number of columns/fields: 19 Columns
- Source and Target Data types
- Grain: Each individual arrest record

Column Name	Source Data Type	Target Data Type (SQL)
ARREST_KEY	String	VARCHAR
ARREST_DATE	V_String	DATE
PD_CD	V_String	INT
PD_DESC	V_String	VARCHAR
KY_CD	String	INT
OFNS_DESC	String	VARCHAR
LAW_CODE	String	VARCHAR
LAW_CAT_CD	String	VARCHAR
ARREST_BORO	V_String	VARCHAR
ARREST_PRECINCT	String	INT
JURISDICTION_CODE	V_String	INT
AGE_GROUP	V_String	VARCHAR
PERP_SEX	String	VARCHAR
PERP_RACE	String	VARCHAR
X_COORD_CD	V_String	INT
Y_COORD_CD	V_String	INT
Latitude	V_String	FLOAT
Longitude	V_String	FLOAT

New Georeferenced Column	String	VARCHAR

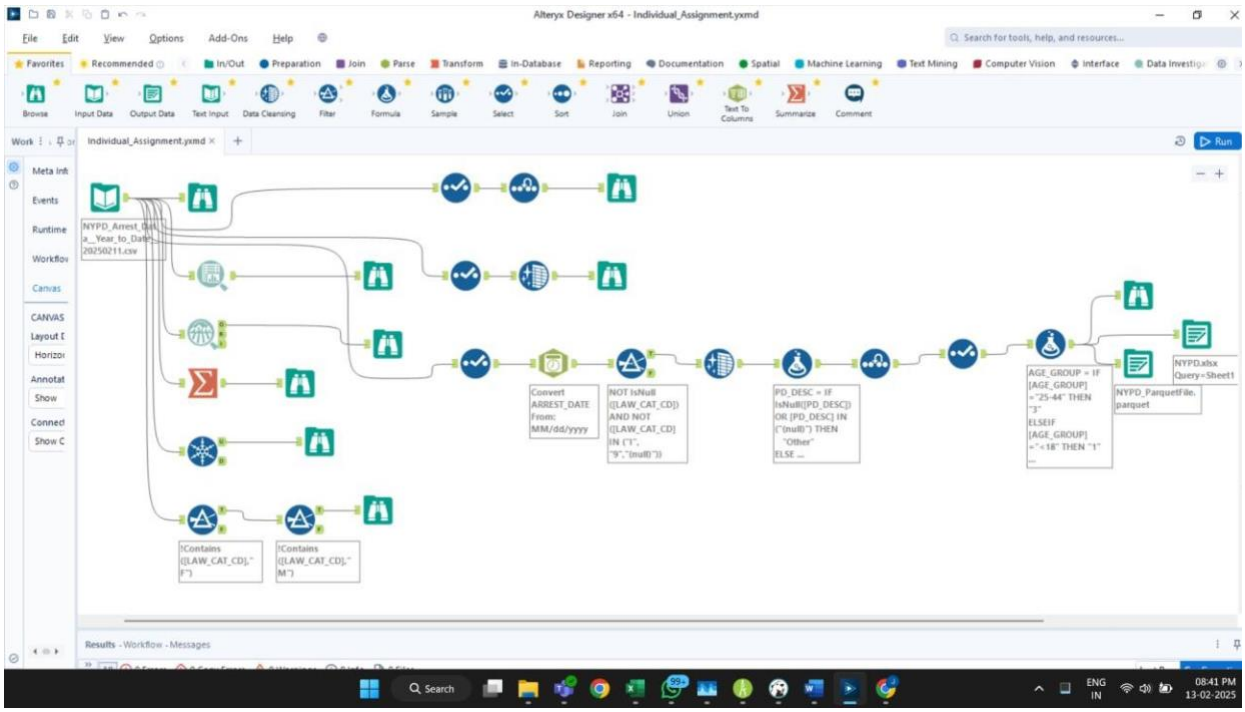
Data Quality Assessment and Data Cleaning

Sno.	Column Name	Issue	Details	Action Plan to fix it
1.	ARREST_DATE	Date in String format	Need to convert it to date format	Use parse DateTime to change the format to DD-MM-YYYY
2.	LAW_CAT_CD	Only 3 levels of Felony present	Other types or Misc felonies reported like "9", "I", "(null)"	Used filter Tool to remove unwanted rows
3.	ARREST_DATE, PD_CD, KY_CD,	Null Values present	Contains Null values in between	Used Data cleaning tool to remove the null values and replace with "0"
4.	PD_DESC	Null Values present	Contains Null values like "(null)"	Used formula tool to replace "(null)" to "Other"
5.	Longitude and Latitude	Null Values	Contains Null values	Used Data Imputation to replace null values with Mean
6.	Select Tool	Wrong Data format	Need to change Data formats	Used Select tool to change the data types
7.	AGE_GROUP	Unusable age group divisions	Need to remove divisions and use some unique identifying values	Used Formula tool to divide age groups into values
8.	LAW_CAT_CD	Jurisdiction out of range	Only 0,1,2 Jurisdictions should be present	Use filter tool to separate rows which are above 2

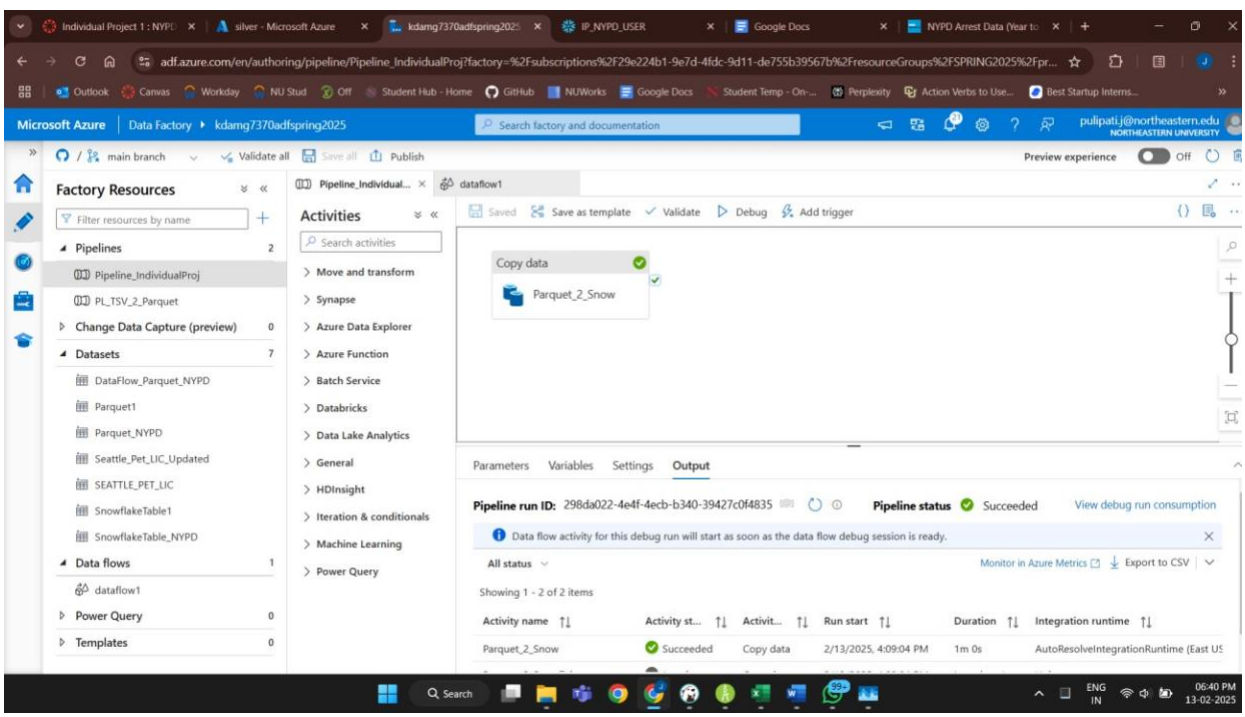
Insights and Observations

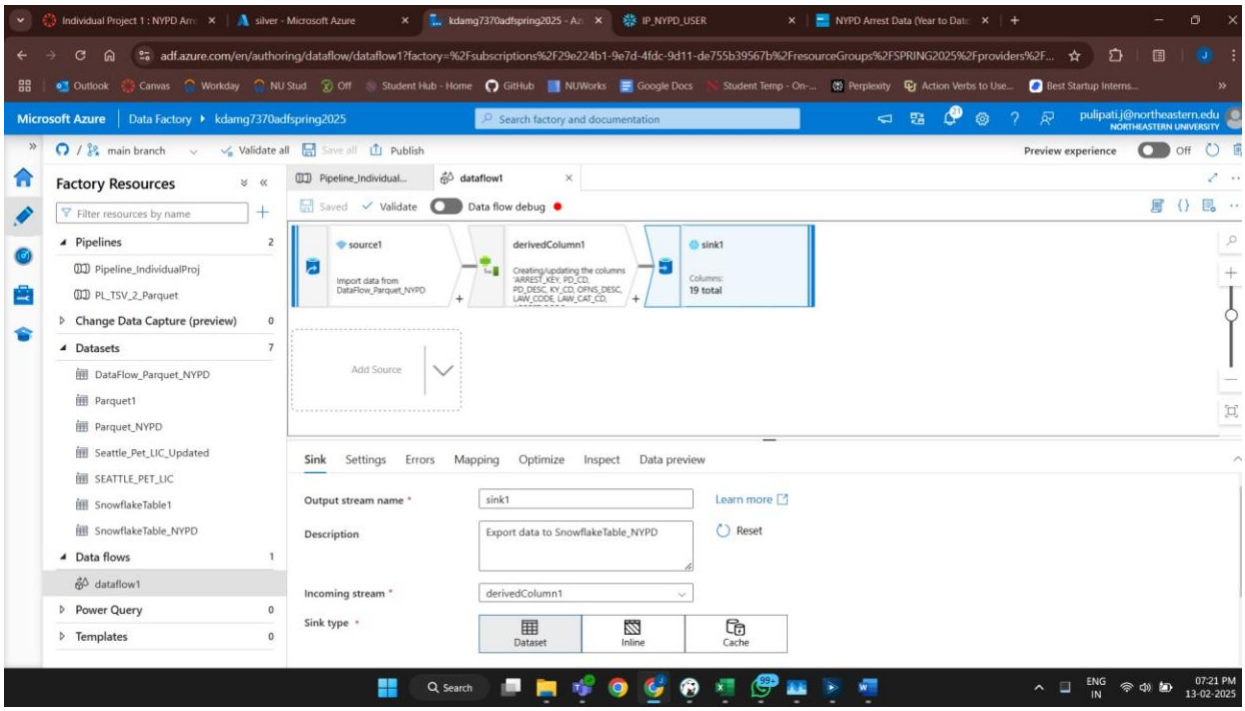
1. The LAW_CAT_CD column has violations other than M, F, V.
2. There are some Jurisdiction codes which are outside the NYPD which can be removed
3. Null values are identified using the Summarise tool
4. There are no duplicate Arrest_key records
5. Some longitudes, latitudes have "0" so used data imputation to fill with average values so that they won't be affected by becoming outliers
6. Converted Age_groups into categories like 1,2,3,4,5.
7. Some column names are not in proper stanndart format, so converted all column names into Capital with no spaces.

Alteryx Workflow

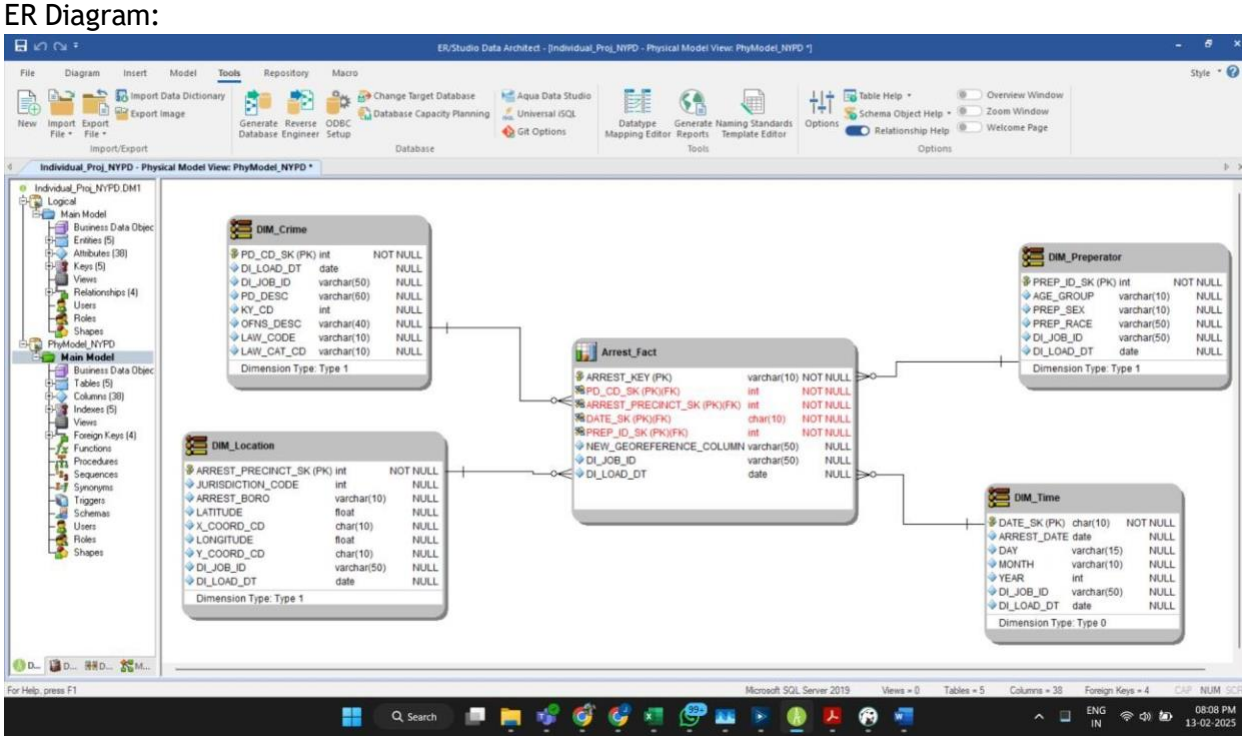


ADF-Snowflake Pipeline

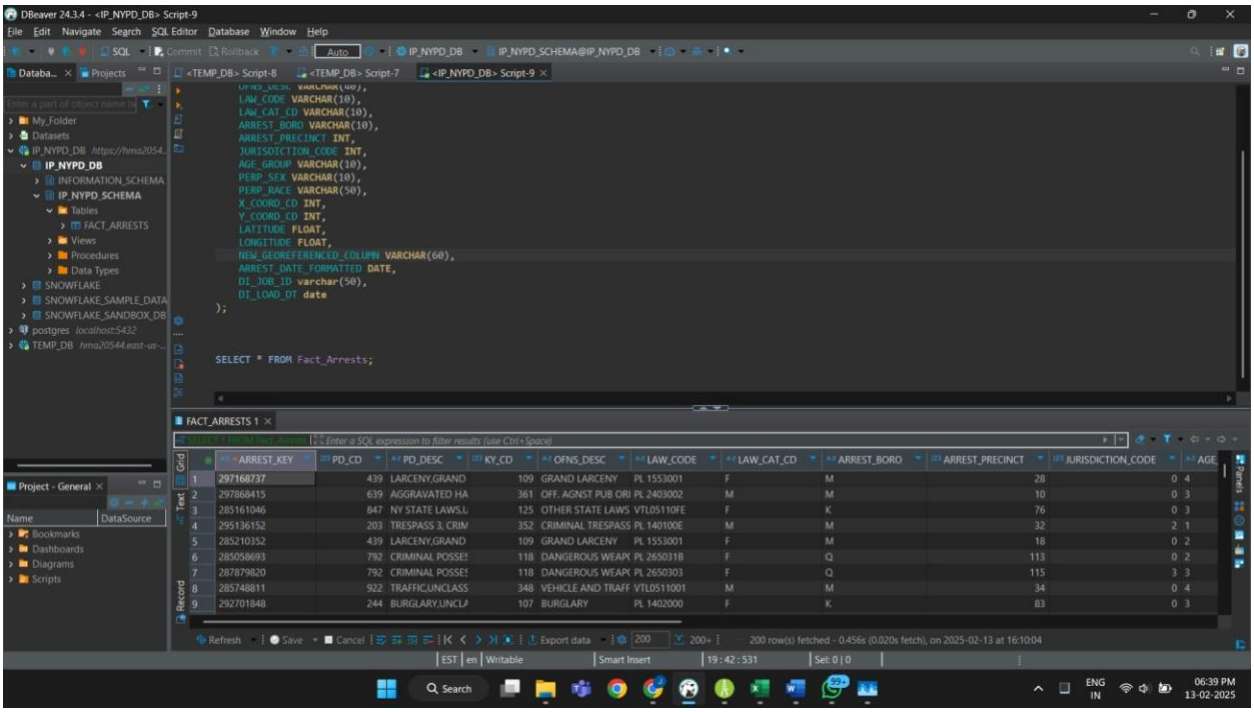



Microsoft Azure
Data Factory
kdamg7370adfspring2025
Factory Resources
Pipelines
Pipeline_IndividualProj
PL_TSV_2_Parquet
Change Data Capture (preview)
Datasets
DataFlow_Parquet_NYPD
Parquet1
Parquet_NYPD
Seattle_Pet_LIC_Updated
SEATTLE_PET_LIC
SnowflakeTable1
SnowflakeTable_NYPD
Data flows
dataflow1
Power Query
Templates
source1
derivedColumn1
sink1
Add Source
Sink
Settings
Errors
Mapping
Optimize
Inspect
Data preview
Output stream name *
sink1
Learn more
Description
Export data to SnowflakeTable_NYPD
Reset
Incoming stream *
derivedColumn1
Sink type *
Dataset
Inline
Cache

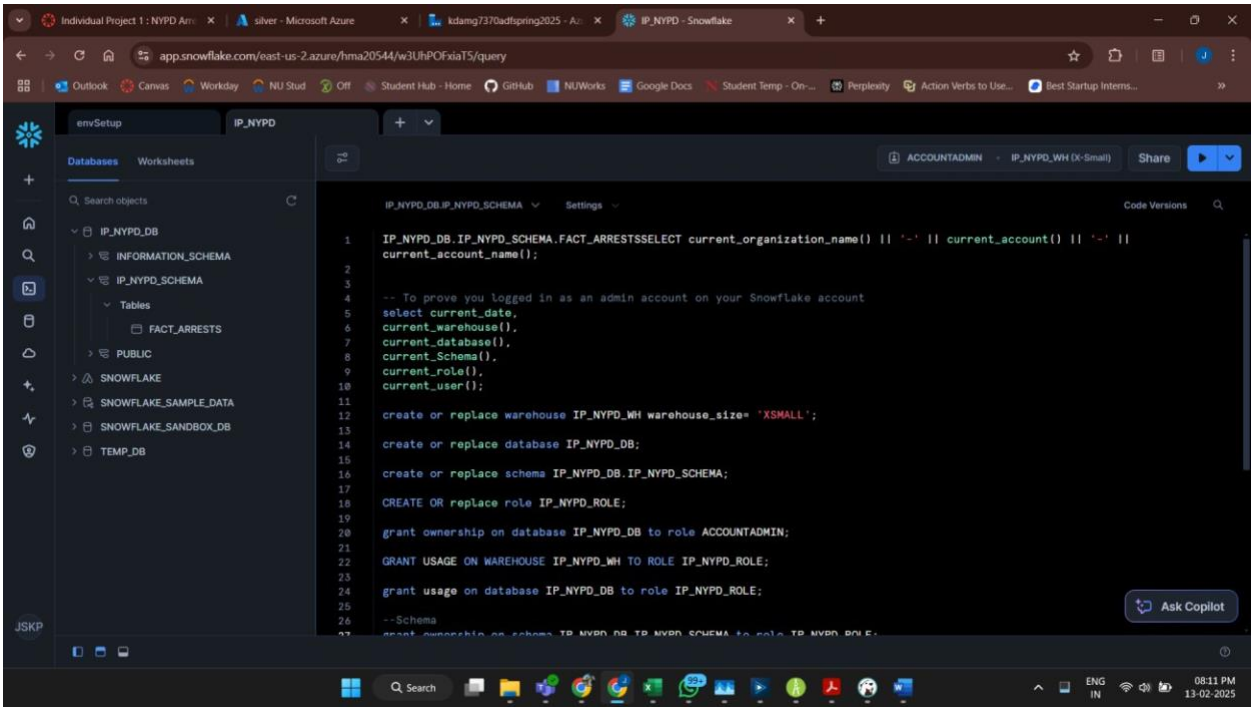
ER Diagram:


DIM_Crime
PD_CD_SK (PK) int NOT NULL
DI_LOAD_DT date NULL
DI_JOB_ID varchar(50) NULL
PD_DESC varchar(60) NULL
KY_CD int NULL
OFNS_DESC varchar(40) NULL
LAW_CODE varchar(10) NULL
LAW_CAT_CD varchar(10) NULL
Dimension Type: Type 1
DIM_Preperator
PREP_ID_SK (PK) int NOT NULL
AGE_GROUP varchar(10) NULL
PREP_SEX varchar(10) NULL
PREP_RACE varchar(50) NULL
DI_JOB_ID varchar(50) NULL
DI_LOAD_DT date NULL
Dimension Type: Type 1
Arrest_Fact
ARREST_KEY (PK) varchar(10) NOT NULL
PD_CD_SK (PK)(FK) int NOT NULL
ARREST_PRECINCT_SK (PK)(FK) int NOT NULL
DATE_SK (PK)(FK) char(10) NOT NULL
PREP_ID_SK (PK)(FK) int NOT NULL
NEW_GEOREFERENCE_COLUMN varchar(50) NULL
DI_JOB_ID varchar(50) NULL
DI_LOAD_DT date NULL
DIM_Location
ARREST_PRECINCT_SK (PK) int NOT NULL
JURISDICTION_CODE int NULL
ARREST_BORO varchar(10) NULL
LATITUDE float NULL
X_COORD_CD char(10) NULL
LONGITUDE float NULL
Y_COORD_CD char(10) NULL
DI_JOB_ID varchar(50) NULL
DI_LOAD_DT date NULL
Dimension Type: Type 1
DIM_Time
DATE_SK (PK) char(10) NOT NULL
ARREST_DATE date NULL
DAY varchar(15) NULL
MONTH varchar(10) NULL
YEAR int NULL
DI_JOB_ID varchar(50) NULL
DI_LOAD_DT date NULL
Dimension Type: Type 0

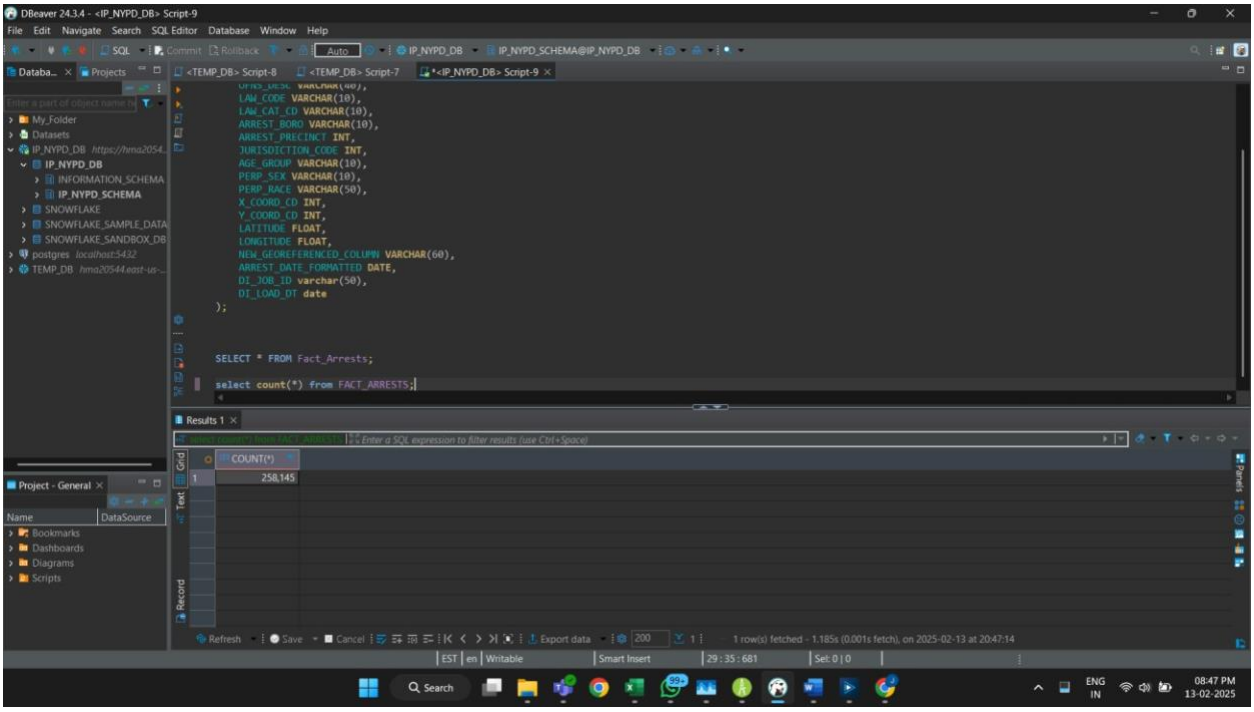
Final Created Table pushed to Snowflake:



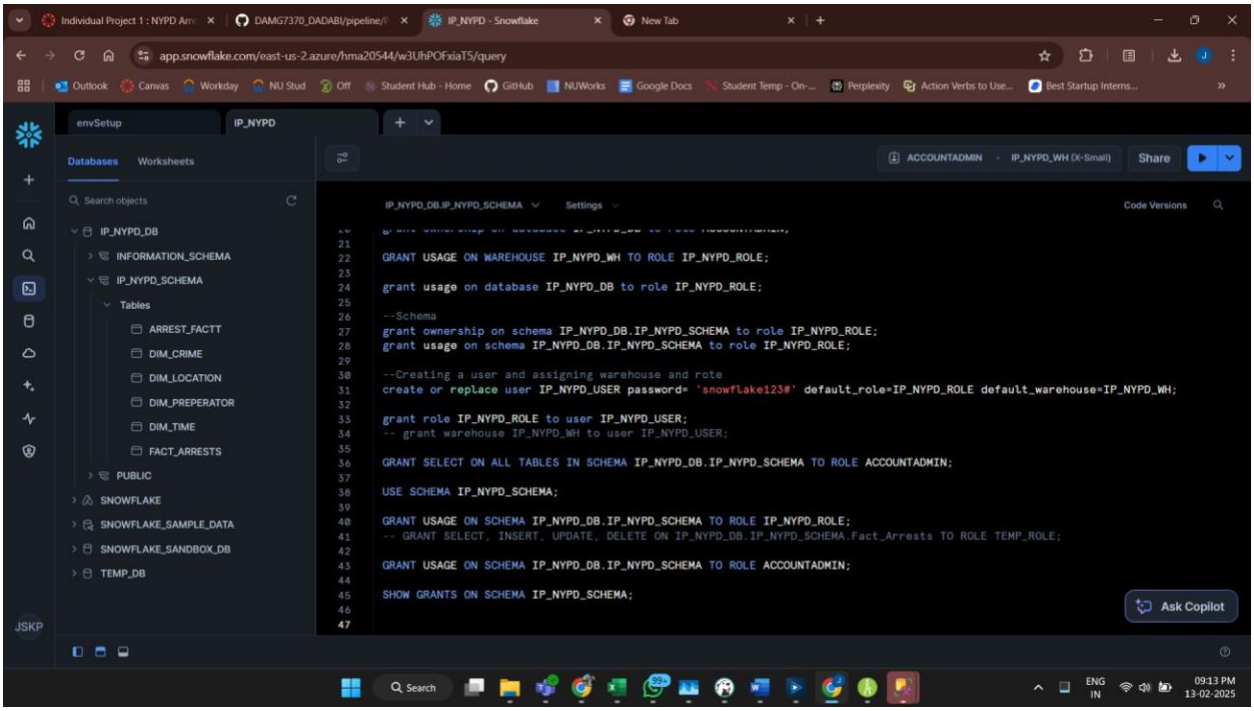
Snowflake DB Creation

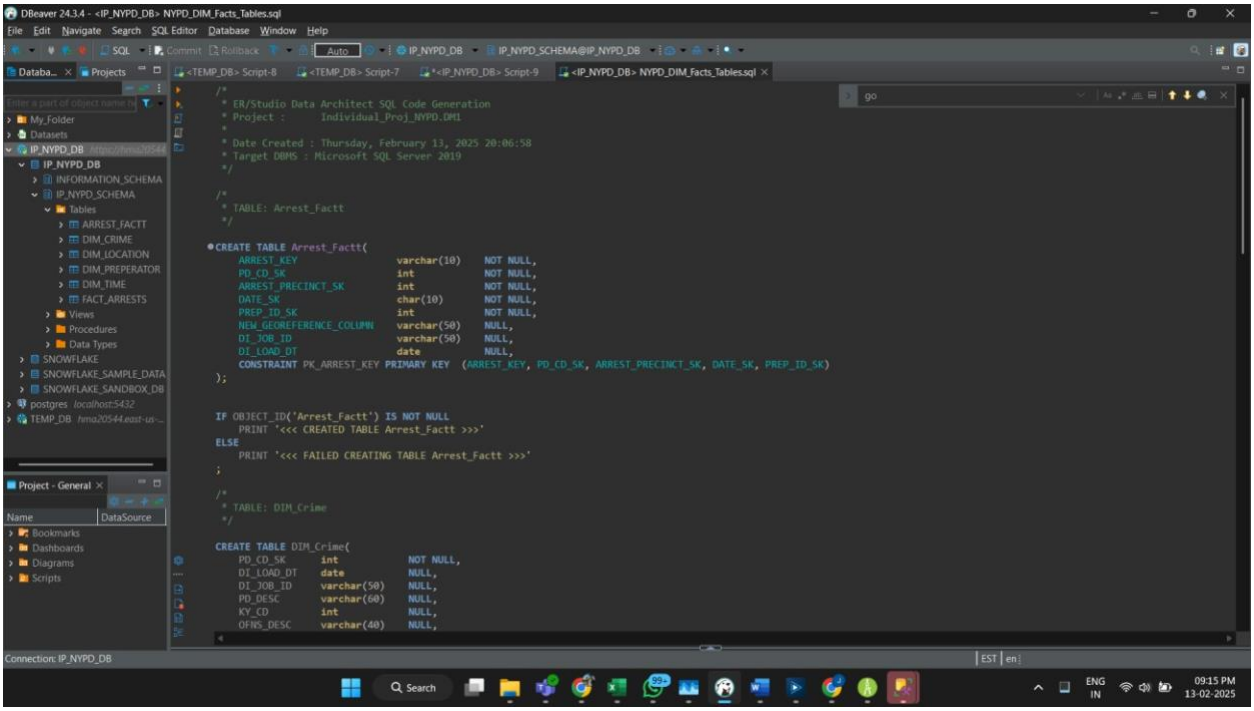


Count (*) rows from Table



DIMENSIONS AND FACTS TABLES:





SQL SCRIPT

```
/*
* ER/Studio Data Architect SQL Code Generation
* Project :     Individual_Proj_NYPD.DM1
*
* Date Created : Thursday, February 13, 2025 20:06:58
* Target DBMS : Microsoft SQL Server 2019
*/


/*
* TABLE: Arrest_Fact
*/


CREATE TABLE Arrest_Fact(
   ARREST_KEY            varchar(10)  NOT NULL,
   PD_CD_SK              int          NOT NULL,
   ARREST_PRECINCT_SK    int          NOT NULL,
```

```
   DATE_SK              char(10)     NOT NULL,
   PREP_ID_SK             int         NOT NULL,
   NEW_GEOREFERENCE_COLUMN    varchar(50)    NULL,
   DI_JOB_ID             varchar(50)  NULL,
   DI_LOAD_DT             date        NULL,
   CONSTRAINT PK_ARREST_KEY PRIMARY KEY NONCLUSTERED (ARREST_KEY, PD_CD_SK,
ARREST_PRECINCT_SK, DATE_SK, PREP_ID_SK)
)

go



IF OBJECT_ID('Arrest_Fact') IS NOT NULL
   PRINT '<<< CREATED TABLE Arrest_Fact >>>'
ELSE
   PRINT '<<< FAILED CREATING TABLE Arrest_Fact >>>'
go

/*
 * TABLE: DIM_Crime
 */

CREATE TABLE DIM_Crime(
   PD_CD_SK      int         NOT NULL,
   DI_LOAD_DT   date        NULL,
   DI_JOB_ID    varchar(50)  NULL,
   PD_DESC       varchar(60)  NULL,
   KY_CD       int         NULL,
   OFNS_DESC      varchar(40)   NULL,
```

```
   LAW_CODE      varchar(10)   NULL,
   LAW_CAT_CD  varchar(10)  NULL,
   CONSTRAINT PK2 PRIMARY KEY NONCLUSTERED (PD_CD_SK)
)

go


IF OBJECT_ID('DIM_Crime') IS NOT NULL
   PRINT '<<< CREATED TABLE DIM_Crime >>>'
ELSE
   PRINT '<<< FAILED CREATING TABLE DIM_Crime >>>'
go

/*
 * TABLE: DIM_Location
 */

CREATE TABLE DIM_Location(
   ARREST_PRECINCT_SK   int         NOT NULL,
   JURISDICTION_CODE      int         NULL,
   ARREST_BORO         varchar(10)  NULL,
   LATITUDE          float       NULL,
   X_COORD_CD          char(10)     NULL,
   LONGITUDE          float       NULL,
   Y_COORD_CD          char(10)     NULL,
   DI_JOB_ID         varchar(50)  NULL,
   DI_LOAD_DT         date        NULL,
   CONSTRAINT PK_ARREST_PRECINCT PRIMARY KEY NONCLUSTERED (ARREST_PRECINCT_SK)
```

```
)

go


IF OBJECT_ID('DIM_Location') IS NOT NULL
   PRINT '<<< CREATED TABLE DIM_Location >>>'
ELSE
   PRINT '<<< FAILED CREATING TABLE DIM_Location >>>'
go

/*
 * TABLE: DIM_Preperator
 */

CREATE TABLE DIM_Preperator(
   PREP_ID_SK    int         NOT NULL,
   AGE_GROUP     varchar(10)  NULL,
   PREP_SEX      varchar(10)   NULL,
   PREP_RACE     varchar(50)  NULL,
   DI_JOB_ID    varchar(50)   NULL,
   DI_LOAD_DT   date        NULL,
   CONSTRAINT PK4 PRIMARY KEY NONCLUSTERED (PREP_ID_SK)
)

go


IF OBJECT_ID('DIM_Preperator') IS NOT NULL
```

```
   PRINT '<<< CREATED TABLE DIM_Preperator >>>'
ELSE
   PRINT '<<< FAILED CREATING TABLE DIM_Preperator >>>'
go

/*
 * TABLE: DIM_Time
 */

CREATE TABLE DIM_Time(
   DATE_SK      char(10)     NOT NULL,
   ARREST_DATE   date        NULL,
   DAY          varchar(15)   NULL,
   MONTH        varchar(10)  NULL,
   YEAR         int         NULL,
   DI_JOB_ID     varchar(50)  NULL,
   DI_LOAD_DT    date        NULL,
   CONSTRAINT PK_DATE_ID PRIMARY KEY NONCLUSTERED (DATE_SK)
)

go


IF OBJECT_ID('DIM_Time') IS NOT NULL
   PRINT '<<< CREATED TABLE DIM_Time >>>'
ELSE
   PRINT '<<< FAILED CREATING TABLE DIM_Time >>>'
go
```

```
/*
 * TABLE: Arrest_Fact
 */

ALTER TABLE Arrest_Fact ADD CONSTRAINT RefDIM_Crime1
   FOREIGN KEY (PD_CD_SK)
   REFERENCES DIM_Crime(PD_CD_SK)
go

ALTER TABLE Arrest_Fact ADD CONSTRAINT RefDIM_Location2
   FOREIGN KEY (ARREST_PRECINCT_SK)
   REFERENCES DIM_Location(ARREST_PRECINCT_SK)
go

ALTER TABLE Arrest_Fact ADD CONSTRAINT RefDIM_Time3
   FOREIGN KEY (DATE_SK)
   REFERENCES  DIM_Time(DATE_SK)
go

ALTER TABLE Arrest_Fact ADD CONSTRAINT RefDIM_Preperator4
   FOREIGN KEY (PREP_ID_SK)
   REFERENCES DIM_Preperator(PREP_ID_SK)
go
```

DDL Scripts

1. How many arrests occurred on any specific day, week, month, quarter, or year?

```
SELECT dt.YEAR, dt.MONTH, COUNT(*) AS Total_Arrests FROM Arrest_Fact af

JOIN DIM_Time dt ON af.DATE_SK = dt.DATE_SK GROUP BY dt.YEAR, dt.MONTH

ORDER BY dt.YEAR, dt.MONTH;
```

2. What are the peak days and months for arrests?

```
SELECT dt.MONTH, COUNT(*) AS Arrest_Count FROM Arrest_Fact af

JOIN DIM_Time dt ON af.DATE_SK = dt.DATE_SK GROUP BY dt.MONTH ORDER BY Arrest_Count DESC;
```

3. What are the top 5 most frequently occurring crimes?

```
SELECT dc.OFNS_DESC, COUNT(*) AS Crime_Count

FROM Arrest_Fact af JOIN DIM_Crime dc ON af.PD_CD_SK = dc.PD_CD_SK

GROUP BY dc.OFNS_DESC ORDER BY Crime_Count DESC;
```

4. Which crimes have increased or decreased the most over time?

```
SELECT dt.YEAR, dc.OFNS_DESC, COUNT(*) AS Arrest_Count FROM Arrest_Fact af JOIN DIM_Time dt ON af.DATE_SK =
dt.DATE_SK JOIN DIM_Crime dc ON af.PD_CD_SK = dc.PD_CD_SK GROUP BY dt.YEAR, dc.OFNS_DESC ORDER BY
dt.YEAR, Arrest_Count DESC;
```

5. Are there specific precincts with higher felony arrests compared to misdemeanors? (Hint: A precinct is a police district within a city.)

```
SELECT dl.ARREST_PRECINCT, dc.LAW_CAT_CD, COUNT(*) AS Arrest_Count

FROM Arrest_Fact af JOIN DIM_Location dl ON af.ARREST_PRECINCT_SK = dl.ARREST_PRECINCT_SK

JOIN DIM_Crime dc ON af.PD_CD_SK = dc.PD_CD_SK

WHERE dc.LAW_CAT_CD IN ('F', 'M') GROUP BY dl.ARREST_PRECINCT, dc.LAW_CAT_CD ORDER BY Arrest_Count;
```

6. Which borough has the highest number of arrests? (Hint: A borough is a large administrative division in NYC, such as Manhattan (M), Brooklyn (K), Queens (Q), The Bronx (B), and Staten Island (S).)

```
SELECT dl.ARREST_BORO, COUNT(*) AS Arrest_Count FROM Arrest_Fact af JOIN DIM_Location dl ON
af.ARREST_PRECINCT_SK = dl.ARREST_PRECINCT_SK

GROUP BY dl.ARREST_BORO ORDER BY Arrest_Count DESC;
```

7. What is the distribution of arrestees by age, race, and gender?

```
SELECT dp.AGE_GROUP, dp.PREP_RACE, dp.PREP_SEX, COUNT(*) AS Arrest_Count

FROM Arrest_Fact af JOIN DIM_Preperator dp ON af.PREP_ID_SK = dp.PREP_ID_SK

GROUP BY dp.AGE_GROUP, dp.PREP_RACE, dp.PREP_SEX ORDER BY Arrest_Count DESC;
```

8. Can we predict high-crime areas based on past arrest data?

```
SELECT dl.LATITUDE, dl.LONGITUDE, COUNT(*) AS Arrest_Count FROM Arrest_Fact af JOIN DIM_Location dl ON
af.ARREST_PRECINCT_SK = dl.ARREST_PRECINCT_SK GROUP BY dl.LATITUDE, dl.LONGITUDE ORDER BY Arrest_Count
DESC;
```